# COMMUNITY OUTREACH PROJECT PROPOSAL

Digital Literacy, Health & Hygiene, DIY Innovation and AI-Powered Livelihoods

## In Collaboration with Aasraa Trust, Dehradun

Aasraa Trust is a Dehradun-based NGO, established in 2009 under the Bindal bridge, that works with street and slum children and migrant communities across Uttarakhand through education, vocational training, nutrition, health and shelter programmes. This proposal outlines a series of hands-on workshops designed and delivered by Computer Science and Engineering students of IIT Roorkee, addressing digital safety, personal hygiene, DIY engineering (drone-making and biology models), and the use of AI tools to build sustainable livelihoods.

> *"The first step to education is basic literacy — and every child, regardless of where they were born, deserves a chance to dream in the language of science, technology, and opportunity."*

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| **Submitted by** | Team, Computer Science and Engineering Department, IIT Roorkee |
| **Partner NGO** | Aasraa Trust, Dehradun, Uttarakhand |
| **Proposed Focus Areas** | Cybersecurity & Digital Safety · Health & Hygiene · DIY Drone-Making & Biology Models · AI for Livelihoods |
| **Team Size** | 10 student members |
| **Document Type** | Project Proposal (tentative plan, subject to Aasraa Trust's inputs) |

# 1. Problem Statement & Overview

Children growing up in slum and migrant communities in Dehradun rarely get a chance to interact with technology, engineering, or structured health education. Most have never touched a circuit board or been taught how to protect themselves from an online scam, and many are never taught basic hygiene practices as a matter of habit rather than instruction. At the same time, families in these communities often have people with real, marketable creative skills — art, design, tailoring — who have no exposure to the free digital tools that could help them earn more from those skills.

This proposal addresses four connected gaps in one coordinated outreach programme with Aasraa Trust: (a) low digital-safety awareness, leaving children and families vulnerable to online fraud; (b) inconsistent hygiene practices that drive preventable illness; (c) little to no hands-on exposure to engineering, robotics or biology; and (d) a missed opportunity to connect existing creative/vocational skills to AI tools that could turn them into a more reliable source of income.

# 2. Background — Why This Problem Matters

Each of these gaps is not a one-off inconvenience — it plays out at a scale that affects millions of children across India every year, and Aasraa's community in Dehradun sits squarely inside that population.

| ~25 lakh | ~4 lakh | ~1 lakh | 2 lakh+ |
|---|---|---|---|
| cybercrime cases reported in India in 2025 alone, up from ~22.7 lakh in 2024, with financial fraud (phishing, OTP theft, UPI scams) behind over 70% of cases | children under five die every year worldwide from diseases linked to inadequate water, sanitation and hygiene (WHO/UNICEF) | under-five diarrhoeal deaths estimated annually in India, closely tied to poor handwashing and sanitation access (UNICEF) | street children were counted across just 10 Indian cities in the most recent nationwide enumeration, most with no access to structured STEM exposure |

Digital fraud is also shifting from metros into smaller cities and semi-urban populations with lower digital literacy — exactly the demographic Aasraa's older students and their families belong to — making basic cyber-hygiene training immediately relevant rather than theoretical. Meanwhile, on the education side, national data shows well over a million children still identified as out-of-school, with children of migrant labourers, slum dwellers and construction workers consistently over-represented in that count. On the livelihoods side, the same communities that lack digital exposure are also the ones least likely to have heard of, let alone used, free AI tools that could meaningfully speed up or improve existing craft- and design-based work.

Aasraa Trust's own centres in Dehradun sit at the intersection of all four problems described above — which is exactly why we believe a single, coordinated outreach programme (rather than four disconnected sessions) can create disproportionate impact here.

# 3. Our Solution

We propose four hands-on, age-appropriate modules, each designed to be run with little to no prior technical background assumed on the part of the students, and each pairing a short conceptual session with a practical, memorable activity.

## Module 1: Cybersecurity & Digital Safety

*Directly targets the OTP/UPI/phishing fraud patterns driving India's cybercrime numbers.*

- Spotting scams, fake OTP calls, phishing links and fraudulent job/lottery messages.
- Understanding deepfakes and manipulated videos/images, with live demonstrations.
- Safe password habits, app permissions, and responsible social-media use.
- Interactive quiz-based assessment to reinforce key takeaways.

## Module 2: Health & Hygiene

*Targets the handwashing and sanitation gaps behind most preventable childhood illness.*

- Handwashing technique, oral hygiene, and safe drinking-water practices.
- Menstrual hygiene awareness session (conducted sensitively, in age-appropriate groups).
- Nutrition basics and food safety relevant to daily life in the community.
- Hands-on demonstration stations (e.g. glow-germ handwashing demo) for retention.

## Module 3: DIY Engineering: Drone-Making & Biology Models

*Gives students the hands-on STEM exposure most never otherwise receive.*

- Team-based assembly of DIY drone kits — motors, propellers, frame, controller — followed by supervised test flights.
- Basic aerodynamics and electronics explained through the drone build itself.
- Biology taught through physical/craft models (e.g. human heart, cell structure, DNA models) built from low-cost, reusable materials.
- Safety briefing and a cordoned test-flight area for all practical sessions.

### Module 4: AI for Livelihoods

*Connects existing creative skills in the community to free AI tools that can improve income.*

- Case study: a student or artisan skilled in poster/art-making learns to use Gemini and similar AI image tools to generate design drafts, mockups and variations quickly.
- Hands-on session where students prompt an AI tool to create a poster, festival greeting, or small-business flyer.
- Discussion on where such skills apply locally — shop signage, event posters, social-media content for small vendors.
- Basic etiquette and limits of AI-generated content, and how to refine/edit outputs for real use.
- Connects directly to Aasraa's existing vocational tracks (art, block printing) by adding a digital, AI-assisted extension.

## 4. Tentative Working Hours (by Week)

We propose spreading the programme across roughly five to six weeks, so that content development, delivery and follow-up documentation are each given enough time without overloading Aasraa's centre schedule. All timings below are tentative and open to adjustment based on Aasraa's academic calendar.

| Week | Activity | Est. Hours |
|---|---|---|
| Week 1 | NGO coordination, site visit, needs assessment with Aasraa staff, plan finalisation | 6 hrs |
| Week 2 | Content development, kit/material procurement, safety planning, team role assignment | 6 hrs |
| Week 3 | Visit 1 — Cybersecurity & Digital Safety + Health & Hygiene workshop | 7 hrs |
| Week 4 | Visit 2 — DIY drone-making & biology model build, supervised test flights | 7 hrs |
| Week 5 | Visit 3 — AI-for-livelihoods workshop + mentor circles + teacher/caregiver briefing | 7 hrs |
| Week 5–6 | Feedback collection, documentation, and final report | 3 hrs |

**Total tentative engagement: approximately 36 hours per team member over 5–6 weeks (21 hours on-ground across 3 visits, plus preparation and documentation).**

## 5. Tentative Budget

| Item | Details | Est. Cost (■) |
|---|---|---|
| Travel | Local transport for team and materials across 3 visits | 9,000 |
| Food & Refreshments | Meals/snacks for team and student participants on workshop days | 8,500 |
| DIY Drone Kits | Motors, propellers, ESCs, controllers, batteries, frames | 4500 |
| Biology Model Materials | Low-cost craft/model-building supplies for anatomy and cell models | 800 |
| Hygiene Kits | Soap, sanitizer, demo material for hygiene sessions | 500 |
| Printed Worksheets & Posters | Handouts for cybersecurity, hygiene and AI-livelihoods sessions | 400 |
| Science activities | Science kits | 2000 |
| **Total** | | 25,700 : 2570 per student Leader |

All figures are indicative planning estimates based on comparable past outreach activity, and will be revised once exact kit quantities and travel logistics for Aasraa's centres are confirmed.

## 6. Possible Impact

- Fewer children and families falling for OTP, phishing and deepfake-based scams, in a demographic that national data shows is increasingly targeted as digital literacy lags behind digital adoption.
- Better everyday hygiene habits, directly relevant given how much preventable childhood illness in India is still tied to handwashing and sanitation gaps.
- First-hand engineering and biology exposure for children who would otherwise have zero access to a robotics lab or science demonstration kit.
- A practical bridge from an existing craft skill (art, poster-making, design) to AI-assisted tools that can make that skill more employable.
- A documented, repeatable partnership model between IIT Roorkee's CSE department and Aasraa Trust that future student cohorts can continue and expand.

*“Dreams transform into thoughts, and thoughts result in action.” — Dr. A.P.J. Abdul Kalam. If even a handful of Aasraa's children walk away from these workshops safer online, healthier, and a little more confident that technology is something they can build rather than just consume, this outreach will have done its job.*

## 7. Tentative Roles of Each Team Member

Roles below are a starting proposal for how the team would divide responsibility across the four modules, logistics, and documentation. These are flexible and will be refined once the plan is finalised with Aasraa Trust.

| # | Name | Roll No. | Tentative Role |
|---|---|---|---|
| 1 | Krishna Gupta | 24114053 | Overall coordination with Aasraa Trust; leads the Cybersecurity & Digital Safety module. |
| 2 | Ish Reyosh Ojha | 24112055 | Co-leads Cybersecurity module; designs the scam/deepfake-spotting quiz and demo material. |
| 3 | Bhavy Ratnman | 24114021 | Leads the Health & Hygiene module; coordinates hygiene-kit procurement and demo stations. |
| 4 | Mahak Khushiramani | 24114057 | Co-leads Health & Hygiene module, including the menstrual-hygiene awareness session. |
| 5 | Aanvi Jha | 24114002 | Documentation and feedback-form design; supports the Hygiene module and final report compilation. |
| 6 | Krish | 24114051 | Leads DIY Drone-Making; handles kit assembly guidance and supervised test-flight safety. |
| 7 | Nitin Yadav | 24114062 | Co-leads Drone module; manages hardware procurement and drone-kit troubleshooting. |
| 8 | Rishit Dhanetwal | 24114079 | Leads the Biology Models track; designs low-cost anatomy/cell-model activities. |
| 9 | Nakshatra Sihotiya | 24114061 | Leads AI for Livelihoods module; prepares the Gemini poster-making demo and prompts. |
| 10 | Abhiraj Bharangar | 24117002 | Co-leads AI for Livelihoods module; manages budget tracking and travel/logistics coordination. |

### Team Contact Details

| Name | Contact | Email |
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We look forward to Aasraa Trust's guidance in adapting this proposal to best fit the needs, age groups and schedule of your learning centres, and to beginning this collaboration.